**Team:**

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**Project Title:**

Third Eye Navigation

## I. Introduction (Sarah Thompson)

Third Eye Navigation is a wearable device intended to be used by people who are blind or otherwise visually impaired. It provides assistance while navigating by detecting objects within immediate walking-range of the user and alerting them accordingly when there is an obstacle in their path.

The main goal of Third Eye is to make navigation safer and easier for those who are blind or visually impaired. The device will act as a third eye, seeing potential obstacles that the user may not be able to see or feel with a traditional mobility cane. By alerting the user to these obstacles, the device will help the user stay aware of their surroundings.

The device will consist of the hardware components listed below:

| Component | Purpose |
|---|---|
| ESP32 Microcontroller | Process sensor data & control other modules |
| Ultrasonic Sensor | Object detection |
| Buzzer | Audio feedback (beeping) |
| Vibration motor | Tactile feedback (vibration) |
| Charging board | To charge the device battery |
| Rechargeable battery | Power supply |
| Breadboard | To connect modules to ESP32 |

The ESP32 microcontroller will be programmed using Arduino IDE to process data from the ultrasonic sensor and interact with the other modules accordingly.

## II. Background research

### a. Existing navigation techniques for the visually impaired (Sarah Thompson)

Navigation can be difficult and sometimes dangerous for those who are blind or severely visually impaired, especially when in an unfamiliar or crowded place. However, there are many tools and techniques that may help make navigation easier so that those with vision loss can maintain their independence and travel safely. Many choose to use a mobility cane, which helps them to identify tripping hazards, crosswalks, or tactile paving in their path. Others may have service

animals or human guides to help them avoid obstacles and get from place to place safely.

i. *Walking stick* (Ryan Shaughnessy)
The walking stick is not infallible. It relies on someone finding it the obstacle by tapping, which can be missed. It also relies on someone making wide enough arcs and continuously tapping the entire time while using it, it's too susceptible to human error. It also has little in the way of detecting incoming obstacles if standing still and has a short range, plus only 2-8% of visually impaired people use them.

ii. *Seeing-eye dogs* (Ryan Shaughnessy)
Seeing-eye dogs are much better but still flawed. Dogs are expensive, eating, vet visits, and grooming (which also requires one to be without said dog for a while). Also get denied from restaurants illegally and people will pet disrupting work. Again, human error. Also most retire after 6-8 years after.

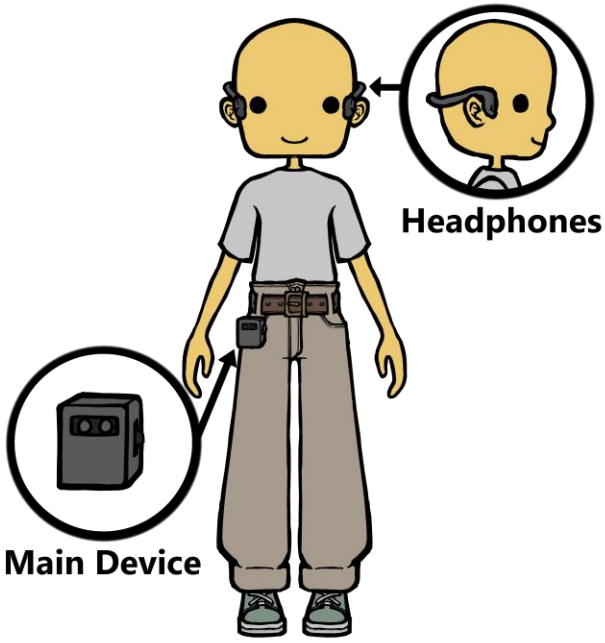
b. <u>What Makes Third Eye Different</u> (Ryan Shaughnessy)

i. Ours isn't as susceptible to human error as these other products. It will vibrate to tell you as long as it continues to send out ultrasonic signals. Plus has a wider range and better "vision".

ii. When you go into the field of computer science, there's one thing they hammer into your head time and time again. Always have backups. And we do, we have backups for data, we have backups for hardware, some of us even have backups for our backups. Now with all these backups we have, why haven't we worked towards allowing a backup for the blind?

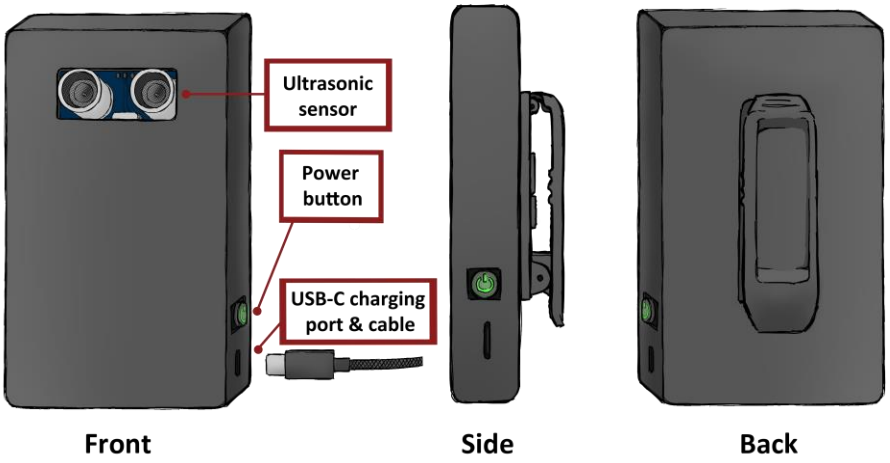
**III. <u>Rough concept diagrams (hand-drawn)</u>** (Sarah Thompson)

a. <u>User Model</u>




i. The main device will contain all hardware enclosed in a plastic case that can be clipped to the user's clothing.
ii. Bluetooth headphones will be optional; users who choose to connect their headphones will be able to receive additional directional cues.

b. <u>Plastic Case</u>




i. The case will be 3D-printed to perfectly fit all hardware components inside.
ii. There will be an opening in the front of the case for the ultrasonic sensor to be able to function; it cannot be obstructed.
iii. There will be a USB-C charging port and a power button.
iv. There will be a belt-clip on the back.

## IV. Hardware components

**a.** Arduino (Sarah Thompson)

i. *Description*

Arduino is an open-source platform used for building electronics projects. It combines hardware and software to be used for a broad variety of purposes. It is extremely versatile and easily accessible to inexperienced beginners and electrical engineers alike.

ii. *Arduino Anatomy*

The primary hardware component is a circuit board that houses a microcontroller and pins. Additional modules, such as sensors or buttons, can be included as needed. The microcontroller is the main processor of the device, handling all of the physical computing. The pins are generally used to handle input or output. A wire will be connected to a pin on one end and a separate module on the other end. The microcontroller will send an electric current through the wire to power the module as dictated by its programming.

iii. *Arduino Programming*

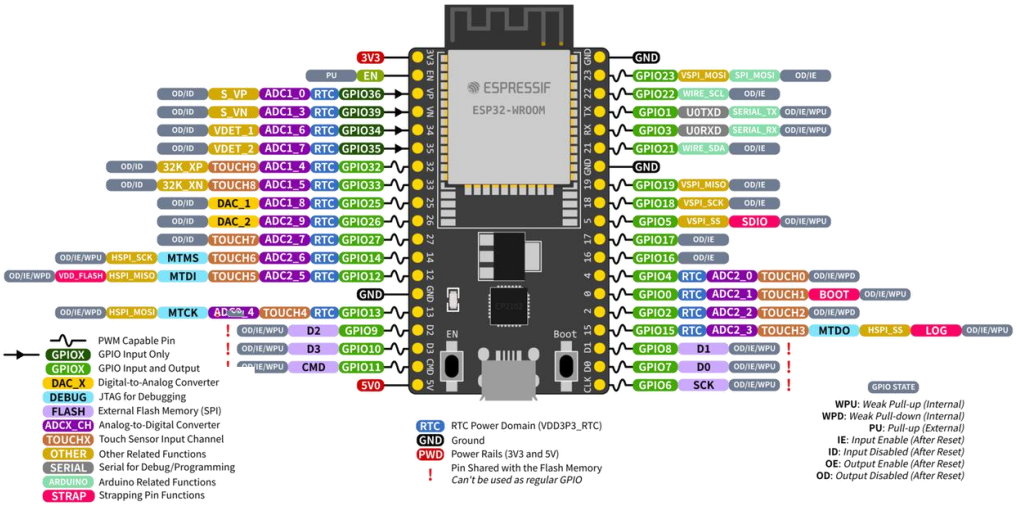
The microcontroller is programmed using a simple programming environment called Arduino IDE. Supported programming languages include C, C++, and assembly. Instructions written through Arduino IDE are sent to the microcontroller on the board, which then processes and computes accordingly. Arduino IDE is also compatible with several other types of microcontrollers, such as the ESP32, which we will be using for this project.

b. ESP32 (Jennifer Greene)

i. There are several physical components that are being used for this hardware focused project. Our microcontroller is the ESP32, which is a low-cost, low-power board that, unlike many basic Arduino boards, has built-in Wi-Fi and Bluetooth capability. This is significant for our project because it means our design can incorporate wireless communication without needing additional hardware. The ESP32 also offers more processing power and memory than entry-level Arduino boards like the Uno, which allows it to handle multiple sensors and more complex logic simultaneously.

ii. An ESP32 board, and other Arduino development boards, all have pins that allow external components to be connected to it. These pins can be inserted into a breadboard, or hardware can be directly connected or soldered onto these pins with wires or just direct connection. These pins

have a connection to each part of the development board to be able to access each part of the development board and the modules that come with it.

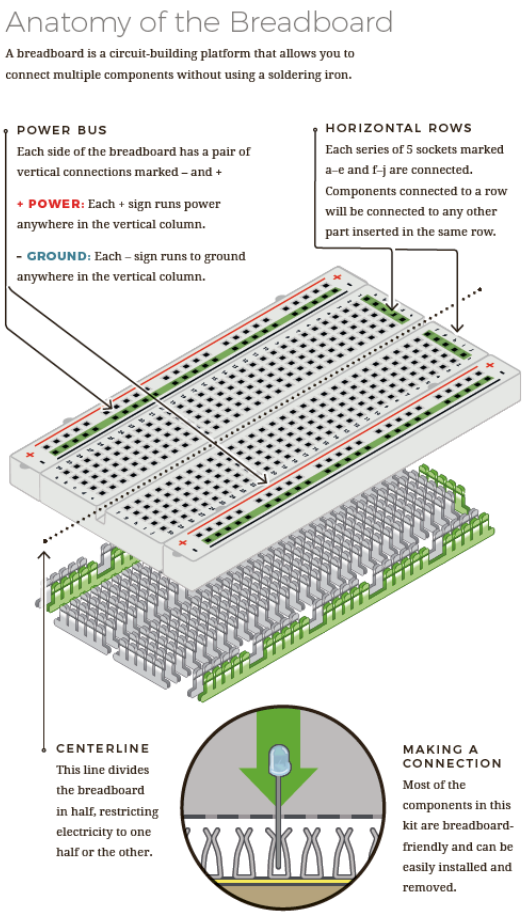
iii. 


c. <u>Breadboard</u> (Jennifer Greene)

i. To connect all our components together, we will use a breadboard. I breadboard is a development board that allows you to connect all of your components without soldering. It makes your project solderless. There are pin holes across the board with rows of solder underneath to connect all the components and send electricity to each. This is going to be used for our proof of concept/demo version of this project. Final products don't use breadboards; they use circuit boards that all the pieces are soldered onto.

1. Breadboards have metal lanes inside that allow for currents to flow between components. Boards have rows labeled with numbers and columns labeled in letters. When placing a component on the board, any pin that is going to relate to a wire to the microcontroller needs to be connected in the same row. Each row has metal lanes. There are special lanes that run perpendicular to the edge of the board called the power bus. They allow you to connect multiple components to the voltage (5V, 3.3V, or VIN) pins and the ground (GND) pins. The voltage bus is usually labeled with a plus sign and in red, while the ground bus is labeled in blue with a negative sign. All components are connected to power and ground.

2.




d. <u>Voltage/Battery</u> (Jennifer Greene)

i. Voltage is the electrical potential difference that drives current through a circuit, essentially the "push" that moves electricity from the power source to each component. Every component in our design (sensors, MCU, motor) has a required operating voltage, and mismatches can damage components or cause them not to function. This is relevant to our project because our components need to share a compatible voltage (most of our parts run on 3.3V or 5V), which affects how we wire the battery, the ESP32, and each sensor/actuator together.

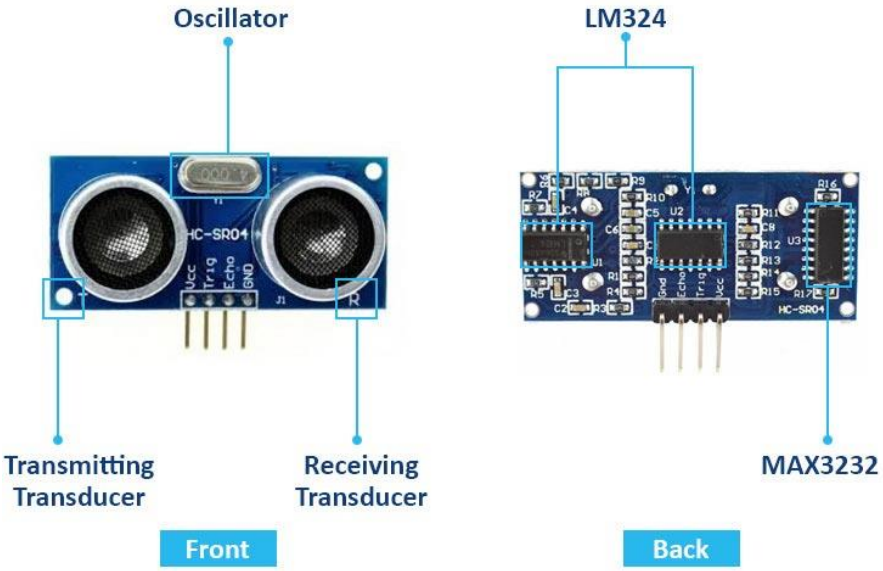
ii. When researching batteries to find what would be compatible with the circuit, we found numerous different batteries. We needed to find a battery that is compact to fit in the product, and the correct voltage to not destroy the microcontroller or our hardware. The ESP32 has two pins allocated for voltage. 3.3V and 5V. These pins indicate the maximum voltage these pins support. A battery at that voltage or less works for

both pins at their respective maximum. The most common battery for Arduino projects is lithium-ion batteries.

e. Sensors (Jennifer Greene)

*i. Ultrasonic*

1. An Ultrasonic Sensor is a component that emits a sound pulse, measures time for the echo to return, and calculates distance to determine how close an object is to the sensor. It uses high-frequency sound waves to measure the distance of objects that may be in the path of the sensor. At its core is the transducer, typically made of a piezoelectric crystal, which both emits and receives ultrasonic waves. The range of this sensor is 2 centimeters (cm) up to 450 centimeters (4.5 meters). This type of component is inexpensive and compact, which is why we are using this for our proof of concept.



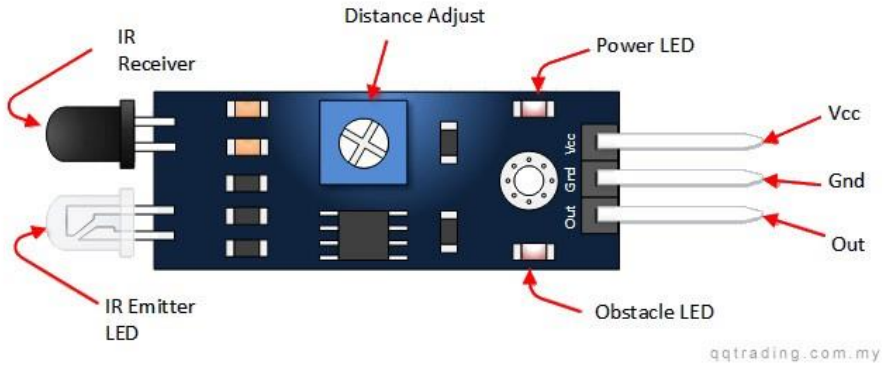

2.

*ii. LiDAR*

1. LiDAR, which stands for Light Detection and Ranging, is a remote sensing technology that uses laser beams to measure precise distances and movement in an environment, in real time. LiDAR works similarly in principle to ultrasonic sensing, but uses pulses of laser light instead of sound, timing the light return to calculate distance. LiDAR offers longer range and more precise, narrower beams than ultrasonic sensors, but are significantly more expensive and often larger. For proof of concept, the expense and

the size is why we aren't using this type of sensor unless ultrasonic is not able to do what we need it to.

iii. *Infrared*

1. An Infrared Detection Sensor is a sensor that uses IR (infrared reflection) to detect obstacles that are in front of the sensor. This works with a led reflecting light into an IR sensor. This works very similar to the ultrasonic sensor, but with light like LiDAR. The difference is that it only can detect that there is an obstacle, but not the distance or where exactly it is. This is a reason why we will not use infrared but research it as an option.
2. 


**f.** <u>Feedback/alert system</u> (Sarah Thompson)

i. *Buzzer*

The buzzer module is part of the alert system for this project. There are two types of electronic buzzers – active buzzers and passive buzzers.

1. Active buzzers only require a DC signal to produce sound because they have a built-in oscillator. This means that it can convert a DC signal into an AC signal. These buzzers can only produce a single fixed sound.
2. Passive buzzers do not have a built-in oscillator, so they require an AC signal to make sound. Additionally, they can be programmed to make a wide variety of sounds.

For this project, a passive buzzer would be more suitable. This is because it would be possible to provide a more in-depth alert system. For instance, the sound produced by the buzzer could become an increasingly higher pitch as the user approaches an obstacle, or a certain sequence of notes may be played to signal that the device battery is low. This may make the alert system more descriptive and intuitive for the user.

ii. *Vibration motor*

The vibration motor will be included to provide the user with haptic

feedback in addition to the buzzer. This will be included in the case of the user not being able to hear audio feedback from the buzzer, such as in noisy places. There are three main types of vibration motor commonly used for this purpose – ERM (eccentric rotating mass) motors, LRA (linear resonant actuator) motors, and less traditionally, piezo actuators.

1. ERM motors function by having an unbalanced weight attached to a mechanical shaft. When the motor receives a DC signal, it will begin to spin, shifting the weight attached to it and creating a centrifugal force. This causes a vibration. Additionally, the strength of the vibration can be changed by controlling the voltage.
2. LRA motors function by having a magnetic weight attached to a spring. When it receives an AC signal, the weight will oscillate, or move up and down, creating a vibration.
3. Piezo actuators use the piezoelectric effect to create vibration. This is a property of certain materials in which they generate electricity when under physical stress. There are different configurations of these actuators, but the most common is the piezoelectric bender. This uses two pieces of piezoelectric material configured in the shape of a beam. When it receives an electric signal, this beam bends, creating movement and therefore, vibration. This method of vibration provides greater control of amplitude and the strength of the movement, allowing for more detailed haptic feedback. However, it requires a much higher voltage signal than ERM or LRA motors.

A flat coin shaped ERM vibration motor would be most suitable for this project, as it can be easily configured to the specific frequency and amplitude needed for the haptic feedback. It will have to be paired with a transistor to control the voltage — this will prevent the motor from overloading the GPIO pins on the ESP32 and causing damage.

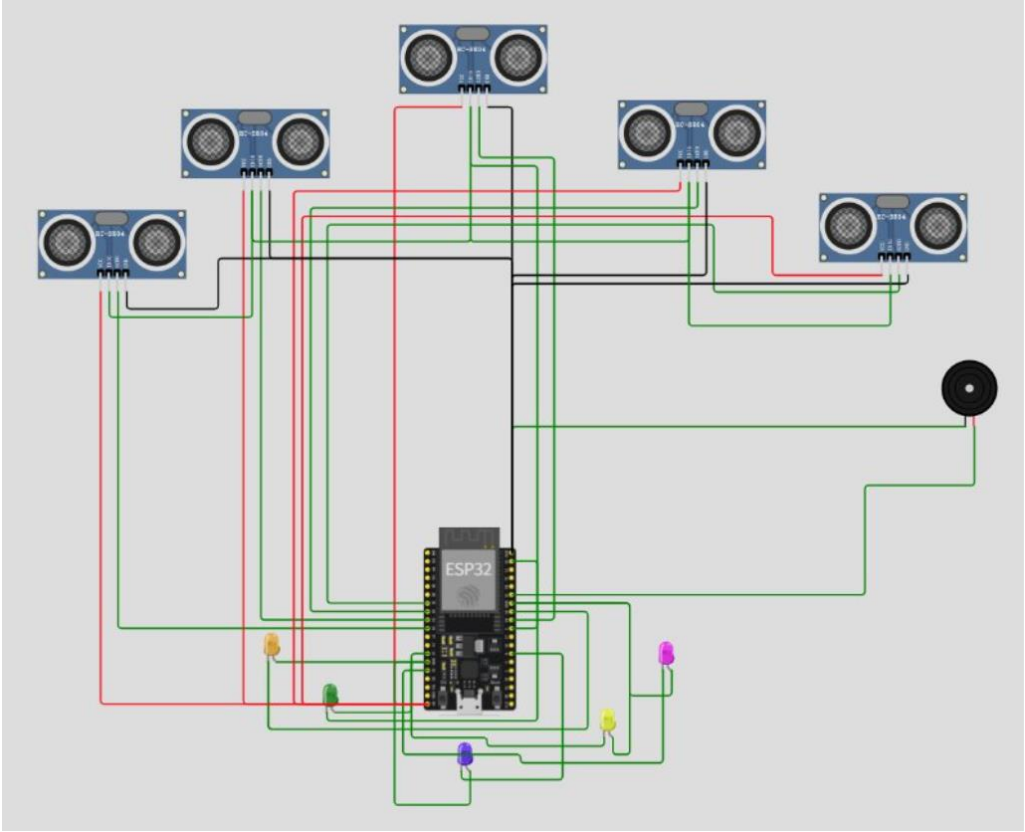
## V. **WOKWI Experiment Page** (YuKun Cai)

a. What is WOKWI?

i. WOKWI is a free open to use online electronics simulator that lets you code, wire, and test microcontroller and IoT projects directly in your web browser. It allows you to simulate popular boards like Arduino (Uno, Mega, Nano), ESP32, STM32, and Raspberry Pi Pico. The simulation allows you to use various components like virtual sensors, LEDs, LCD/OLED displays, motors, buttons, and logic analyzers, along with

writing the code and using the libraries that would be needed to work on the real project.

b. Our experiment
    i. We used this platform to experiment with the components we planned to use to get a better idea of what we were trying to accomplish.
    ii. Firstly, I suggest using five hc-sr04 sensors. This will enable a wider detection range and keep the cost at a very low level.
    iii. Secondly, I used five different colored LED lights to represent the five physical vibrators. These five vibrators (LEDs) correspond one-to-one with the five object detectors mentioned above.
    iv. Finally, there is a buzzer that provides an alert sound to alert the user when there are obstacles in the detection direction.
    v. Working principle: Firstly, the five detectors can respectively detect the left, left middle, middle, right middle and right sides in a total range of 120° from left to right. The corresponding physical vibrators (LEDs) are orange, green, blue, yellow and purple. Currently, the closest detection distance is set at 100 cm (which can be adjusted to 30 cm), and the farthest is approximately 400 to 500 cm. During normal operation, when the detector detects an obstacle within 100 cm, the corresponding physical vibrator (LED) will start to work, and at the same time, the buzzer will emit an alarm sound. When the user passes over the obstacle, the device returns to normal operation.

vi. 


VI. **3D-Printing** (James Pazik)

Computer-aided design (CAD) software lets designers create, edit, and optimize designs as 2D drawings or 3D models. CAD software is used in many fields, including computer science, engineering, architectural design, and manufacturing. Some common attributes of CAD software include the following. The software can be used to design models with precise measurements. Allows parametric design, which saves every step taken in the design process and creates a timeline of changes. The user has the ability to go back to any point on the timeline while preserving the changes that took place afterward. If any change is made in the past, all future changes will be adjusted for it, as if that change always was that way. This is an extremely powerful feature that allows the project's design to be adaptable. After creation of a model, the software can export the model as a 3d-printable .stl file. For this project, the CAD software chosen was Autodesk Fusion 360. The software has all the previously discussed features and is free for students and for personal use.

The purpose of slicing software is to convert the exported .stl file into printable G-Code. G-Code, or Geometric Code, is the language used to control every aspect of the 3D printer, such as directions for printing layers along with all the printing settings. Some of the settings include print speed and temperature, layer height, and where to

place support for the structure. For this project, the selected slicing software is UltiMaker Cura, which is open source and free to use.

## VII. Sources


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